

PERFORMANCE WORK STATEMENT (PWS)

PSNS & IMF Radiography X-ray Systems Maintenance and Repair Services

PART 1 GENERAL INFORMATION

1. **GENERAL:** This is a non-personal services contract to provide maintenance, troubleshooting and repairs services for Government owned X-ray systems utilized to perform radiography at Puget Sound Naval Shipyard. The Government shall not exercise direct supervision or control over any contractor employees. Notwithstanding, contractor shall ensure employees comply with all requirements of this Performance Work Statement (PWS) and contract.

1.1 Description of Services/Introduction: The contractor shall provide all personnel, equipment, supplies, facilities, transportation, tools, materials, supervision, other necessary items and non-personal services to perform maintenance, troubleshooting, and repair services on government-owned X-ray systems at Puget Sound Naval Shipyard, as defined in the Performance Work Statement. The contractor is not responsible for providing items specified as government-furnished property and services. All work must be performed to the standards in this contract.

1.2 Background: The Non-Destructive Testing Division, Code 135, of the Quality Assurance Department at PSNS performs Radiography Testing among other non-destructive inspection methods. Code 135 currently utilizes three X-ray systems to perform Radiography Testing on US Navy Ships piping assemblies and components. The X-ray systems require quarterly maintenance to operate at optimal performance and meet all safety standards during radiography operations. C/135 currently has a contract (N4523A24P5703) with Test Equipment Distributors to perform quarterly maintenance with options for unscheduled services. Due to C/135 replacing two of the existing X-ray systems with Comet iVario 450 systems, C/135 will need to solicit a new contract that will cover maintenance on the new systems as well as the existing Gulmay Cabinet X-ray system. This contract is to solicit a contractor to perform preventive maintenance, troubleshooting and repair services for one Gulmay 320 and two Comet 450 X-ray systems, which are located in the Controlled Industrial Area of Puget Sound Naval Shipyard.

1.3 Objectives:

1.3.1 Ensure comprehensive quarterly preventive maintenance on all X-ray systems to ensure peak performance and full compliance with all federal, state, and manufacturer safety requirements.

1.3.2 Ensure all necessary troubleshooting and corrective maintenance on the X-ray systems to restore full functionality following any equipment malfunction or failure.

1.4 Scope: The contractor shall provide quarterly preventative maintenance, troubleshooting and corrective maintenance services on three radiography X-ray systems located at Puget Sound Naval Shipyard.

1.5 Period of Performance: The ordering period for this contract shall be 5 years from date of contract award.

1.6 Quality Assurance: The government shall evaluate the contractor's performance under this contract in accordance with the Quality Assurance Surveillance Plan. This plan is primarily focused on what the Government must do to ensure that the contractor has performed in accordance with the performance standards. It defines how the performance standards will be applied, the frequency of surveillance, and the minimum acceptable defect rate(s).

1.7 Location:

1.7.1 Place of Performance: The work to be performed under this contract will be at the government facility, which is Puget Sound Naval Shipyard, [REDACTED] NDT Lab Radiography Area.

1.7.2 Hours of Operation: The contractor is responsible for conducting business between the hours of 0720 - 1600 Monday through Friday except Federal holidays or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings. For other than firm fixed price contracts, the contractor will not be reimbursed when the government facility is closed for the above reasons. The Contractor must at all times maintain an adequate workforce for the uninterrupted performance of all tasks defined within this PWS when the Government facility is not closed for the above reasons. When hiring personnel, the Contractor shall keep in mind that the stability and continuity of the workforce are essential.

1.7.3 Recognized Holidays: Contractor is not required to perform services on federal holidays. The list of current federal holidays can be found here: <https://www.opm.gov/policy-data-oversight/pay-leave/federal-holidays/>

In addition to the above listed holidays, Puget Sound Naval Shipyard is normally closed for all except necessary services during the week between Christmas Day and New Year's Day (curtailment). Necessary services/frequencies for various facilities are noted on the detail sheets and work contracted for shall be performed during this period unless otherwise directed by the Contracting Officer. Exact number of closure days may vary from year to year. Actual requirements will be determined in advance of the closure. On site work performed by the Contractor for the Contractor's convenience during this time must also be specifically requested by the Contractor in writing and approved by the Contracting Officer in advance.

1.8 Security Requirements: The contractor shall ensure all employees performing work under this contract obtain and maintain the required badges and access credentials for the government facility located at Puget Sound Naval Shipyard, [REDACTED] NDT Lab Radiography Area. The requirements and procedures are detailed in the "Badging and Access Requirements" attachment to the solicitation/contract.

1.8.1 Physical Security: The contractor shall be responsible for safeguarding all government equipment, information and property provided for contractor use. At the close of each work period, government facilities, equipment, and materials shall be secured.

1.9 Special Qualifications: The Contractor shall ensure all personnel performing preventive maintenance, troubleshooting, and corrective maintenance are authorized manufacturer representatives (AMR) currently certified by both Gulmay and Comet for their respective X-ray systems used for industrial radiography. In addition, the Contractor shall have a minimum of 5 year' experience in preventative maintenance and repair of industrial radiography X-ray systems and be an authorized Comet and Gulmay distributor for OEM parts.

1.10 Post Award Conference/Periodic Progress Meetings: The Contractor agrees to attend any post award conference convened by the contracting activity or contract administration office in accordance with FAR Subpart 42.5. The contracting officer, Contracting Officers Representative (COR), and other Government personnel, as appropriate, may meet periodically with the contractor to review the contractor's performance. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the government.

1.11 Contracting Officer Representative (COR): The COR monitors all technical aspects of the contract and assists in contract administration. The COR is authorized to perform the following functions: assure that the Contractor performs to the technical requirements of the contract; perform inspections necessary in connection with contract performance; maintain written and oral communications with the Contractor concerning technical aspects of the contract; issue written interpretations of technical requirements, including Government drawings, designs, specifications; monitor Contractor's performance and notifies both the Contracting Officer and Contractor of any deficiencies; coordinate availability of government furnished property, and provide site entry of Contractor personnel.

The COR is NOT authorized to change any of the terms and conditions of this contract. **Only the warranted Contracting Officer is authorized to make modifications/changes to the terms and conditions of this contract.** If the Contractor believes any Government personnel has effectively issued a change resulting in a change in effort and price/cost of the contract, the Contractor shall immediately notify the Contracting Officer in writing.

CORS:

PRIMARY
[REDACTED]
[REDACTED]

1.12 **Contract Manager:** The contractor shall provide a contract manager who shall be responsible for the performance of the work. The name of this person and an alternate who shall act for the contractor when the manager is absent shall be designated in writing to the contracting officer. The contract manager or alternate shall have full authority to act for the contractor on all contract matters relating to execution of this contract.

1.12.1 **Identification of Contractor Employees:** All contractor personnel working in situations where their contractor status is not obvious to third parties are required to identify themselves as such to avoid creating an impression that they are Government officials. All contractor personnel on a government installation will be required to obtain and wear badges in the performance of this contract.

1.13 **Safety Requirements:**

1.13.1 The Contractor shall provide the government with a Lockout/Tagout procedure to perform work according to this PWS. The Contractor shall not perform any servicing or maintenance on the X-ray machines and associated systems where unexpected energization, startup, or release of stored energy could occur and cause injury or damage. All equipment shall be isolated in accordance with the Contractor's Lockout/Tagout procedures. Coordinate Lockout/Tagout activities with S/06 Maintenance personnel as necessary. Personnel shall not be considered protected until all hazardous energy control procedures have been fully implemented. Protection from electrical hazards, including arc flash, shall comply with all applicable regulatory requirements and established safety standards.

1.13.2 The Contractor shall submit completed PSNS&IMF 5090/132 CHMI form and Safety Data Sheets as requested by COR for all chemicals to be used during maintenance a minimum of one month prior to first site visit and thereafter if there are any changes to the chemicals used. CHMI and Safety Data Sheets are needed for the Contractor to get authorization from PSNS Code 106 HazMat for chemicals to be brought and used inside PSNS.

1.13.3 The Contractor shall submit a completed PSNS Receipt and Monthly Usage Form to COR for each quarterly visit to report the amounts of chemical originally identified on CHMI that were used to perform quarterly maintenance. This form will be provided by COR to Contractor upon completion of maintenance.

PART 2
DEFINITIONS & ACRONYMS

2. **DEFINITIONS AND ACRONYMS:**

2.1. COMMON DEFINITIONS: Per Clause 52.212-4(e), clause 52.202-1 is incorporated by reference to this contract.

2.2 SPECIAL DEFINITIONS:

CHMI – Contractor hazardous Material Inventory form

NDT – Non-Destructive Testing

OEM – Original Equipment Manufacturer

PSNS – Puget Sound Naval Shipyard

PART 3 GOVERNMENT FURNISHED PROPERTY, EQUIPMENT, AND SERVICES

3. **GOVERNMENT FURNISHED ITEMS AND SERVICES:**

3.1. Services: The Government will provide:

3.1.1 Transportation from the Pass and ID facility or the Bremerton parking lot to the C/135 NDT Lab as necessary

3.1.2 Personnel to escort contractors inside Radiography Area in the C/135 NDT Lab during maintenance and unscheduled visits

3.1.3 Purchase any parts needed by the contractor to complete repairs on X-ray systems

3.1.4 Disposal of hazardous waste generated in the performance of the work as defined in this PWS

3.2 Equipment:

3.2.1 Vault 1 iVario 450 X-ray System consisting of:

1. X-Ray Tube: Comet MXR-451/26, S/N 2175193 (No Asset ID assigned)
2. Power Generators: Comet iVario-450/4.5, S/N 2178494/2177080
3. X-ray Controller: Comet, S/N 04712627
4. Cooler: Comet XRC-4523-OA, S/N 5038825L02
5. Tube Support Stand

3.2.2 Vault 3 iVario 450 X-ray System (Asset 4523A25787) consisting of:

1. X-Ray Tube: Comet MXR-451/26, S/N 2148020
2. Power Generators: Comet iVario-450/4.5, S/N 2114939 and 2099754
3. X-ray Controller: Comet iVario-450/4.5, S/N 04403651
4. Cooler: Comet XRC-4523-OW, S/N 5024434L01
5. Tube Support Stand

3.2.3 Cabinet Gulmay 320 X-ray System (Asset 4523A16049) consisting of:

1. X-Ray Tube: Mfr. Comet, Asset ID: 4523A18189, S/N 1102260
2. Power Generators: Model CP/160/1 S/N 1479-0917/no visible label on other unit
3. X-ray Controller: Model MP1/19 S/N 3991-0917
4. Cooler: Model OL4503, S/N 71829 12L04

PART 4
CONTRACTOR FURNISHED ITEMS AND SERVICES

4. CONTRACTOR FURNISHED ITEMS AND RESPONSIBILITIES:

4.1 General: The Contractor shall furnish all supplies, equipment, facilities and services required to perform work under this contract that are not listed under Section 3 of this PWS.

4.2. Materials: The Contractor shall furnish materials, supplies, and equipment necessary to meet the requirements under this PWS.

4.3. Equipment: The Contractor shall furnish all equipment necessary to perform preventative maintenance, troubleshooting and corrective maintenance in accordance with this PWS.

PART 5
SPECIFIC TASKS

The contractor shall perform quarterly preventative maintenance, as well as unscheduled troubleshooting and corrective maintenance, on the three (3) radiography X-ray systems located at PSNS and listed under paragraph 3.2. These tasks are required to ensure the systems operate at peak performance and maintain full compliance with all federal, state, and manufacturer safety requirements. The Contractor shall perform the following tasks:

5.1 Scheduled Maintenance

A summary of the required checks and services to be performed quarterly.

System	Key Tasks
X-ray System	• Power-up, run diagnostics, and verify X-ray machine warms up to max kV and max mA with no error codes • Test tube power-limiting safety circuits. • Inspect tube head and transformer for security, leaks, or damage. • Check high-voltage safety discharge mechanisms. • Inspect high-tension (HT) cables for proper routing and wear. • Clean and lubricate HT cable connections. • Inspect internal electronics for damage, heat, or loose components.
Safety Systems	• Functionally test all power disconnects and door interlock switches. • Verify all audible/visual pre-exposure warnings and X-Ray OFF controls.
Wiring & Electrical	• Inspect all control wiring and high-tension cables for damage, frays, or stress.
Enclosure & Cabinet	• Inspect cabinets for damage. • Verify all warning signs, beacons, and emergency stop buttons are operational.
Mechanical Systems	• Lubricate all moving assemblies. • Check and adjust all mechanical safety stops and electronic limit switches. • Inspect moving parts for excessive wear or abrasion.

Control Console	• Verify and calibrate all operator control functions (kVp, mA, mAs, etc.).
Cooling System	• Inspect all cooling lines, hoses, and fittings for leaks or degradation. • Verify all cooling safety circuits (flow, temperature, level) are functional. • Change the oil filter if required.
Software	• Check for OEM software updates, install approved patches, and backup system configurations as permitted by PSNS IT Department.

5.2 Administrative & Reporting

Requirement	Description
Labor Allowance	Provide a minimum of 8 direct work-hours per maintenance event (excludes travel, admin and site access time).
Invoicing	Invoice each of the three X-ray machines separately per visit.
Pre-Visit Submissions	Submit required HazMat documents and Lockout/Tagout plans one month prior to first maintenance visit and whenever there are any changes.
Service Report	Submit a detailed service report after each visit, noting any identified issues and recommendations.

5.3 Corrective Maintenance (Repairs)

Actions for unscheduled troubleshooting and repair services.

Requirement	Description
Trigger	Any malfunction including an event preventing the system from powering up to max kV/mA without error codes.
Initial Response	Acknowledge service request within 24 hours.
On-Site Arrival	Technician must be on-site within 7 business days of the service request.
Labor Allowance	Includes 8 hours for on-site diagnosis and repair (excludes travel and site access).
Post-Repair	Technician must test the system to confirm it meets original operating standards.
Parts	Submit a service report with a quote for all necessary parts. The Government will purchase the parts.
Repair Completion	Accomplish repairs within 2 weeks after parts arrive on-site.

5.4 Quality & Compliance

Requirement	Description
Regulatory	All work must comply with federal and state regulations, particularly NRC rules for radiation safety.
Reporting	Service reports must be detailed enough to demonstrate regulatory compliance and track equipment history.

PART 6 APPLICABLE PUBLICATIONS

6. APPLICABLE PUBLICATIONS (CURRENT EDITIONS):

6.1. The Contractor must abide by the Operator Manuals for the Comet iVario 450 and Gulmay 320 X-ray systems.

PART 7 ATTACHMENT/TECHNICAL EXHIBIT LISTING

7. ATTACHMENT/TECHNICAL EXHIBIT LIST:

7.1 Technical Exhibit 1 – Performance Requirements Summary:

7.2. Technical Exhibit 2 – Deliverables Schedule:

7.3 Attachment 1/Technical Exhibit 3 – PSNS&IMF 5090/132 CHMI Form

TECHNICAL EXHIBIT 1

Performance Requirements Summary

The contractor service requirements are summarized into performance objectives that relate directly to mission essential items. The performance threshold briefly describes the minimum acceptable levels of service required for each requirement. These thresholds are critical to mission success.

Performance Objective (The Service required—usually a shall statement)	Standard	Performance Threshold (This is the maximum error rate. It could possibly be “Zero deviation from standard”)	Method of Surveillance
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PRS # 1. Operational System Performance The contractor shall ensure each X-ray system is fully operational and calibrated post-maintenance, with all functions verified. PWS paragraph 5.1, 5.3	System must pass all OEM diagnostic routines with validated output parameters (kVp, mA).	100% operational readiness upon sign-off.	Verify machine warmup with no error codes and service report analysis. Senior Radiographer, [REDACTED]
PRS # 2 Mechanical & Physical Integrity The Contractor shall inspect, service and secure equipment enclosures, mechanical movements, and cooling systems PWS paragraph 5.1	All moving parts are lubricated, secure, and free from excessive wear/leaks. Enclosures are secure and clean.	Zero leaks (oil/water); Zero unsecured components; Smooth operation of mechanical parts.	Visual inspection, mechanical function check, and service report sign-off. Senior Radiographer, [REDACTED]
PRS # 3 Documentation & Software Management The Contractor shall ensure the system software is current, and all service actions fully documented. PWS paragraph 5.1	Required software patches installed (if applicable), system configuration backed up, and a comprehensive service report provided to the COR.	100% of required documentation is provided within 24 hours of service completion.	Review of service reports, software version logs, and backup confirmation files. Senior COR [REDACTED]
PRS # 4 Safety & Regulatory Compliance The contractor shall ensure all safety systems and physical integrity checks are completed and documented. PWS paragraph 5.1	All safety interlocks (doors, E-Stops, warnings, shorting bars) must be functional.	100% compliance with FDA, State, and OEM safety standards; Zero safety failures observed.	Safety checklist verification, radiation survey report review, and spot checks. Radiation leakage must be confirmed below 0.5 mR/hr. Senior Radiographer, [REDACTED]
PRS # 5 Lockout/Tagout (LOTO) Procedures The Contractor shall provide, adhere to, and coordinate LOTO procedures for all servicing where hazardous energy exists. PWS paragraph 1.13.1, 5.2	LOTO procedure must be provided to the government and strictly followed. All work performed safely in accordance with LOTO and electrical safety standards.	100% compliance with established LOTO procedures; Zero safety incidents related to energy control.	LOTO procedure review and safety records review. COR [REDACTED]
PRS # 6 Chemical & Hazardous Material Management The Contractor shall submit required forms (CHMI, SDS, Monthly Usage) for all chemicals used on site. PWS paragraph 1.13.2, 1.13.3	Submissions must be timely (one month prior for initial use) and accurate. All chemicals used must be authorized by PSNS Code 106 HazMat.	100% of required forms submitted accurately and on time; Zero unauthorized chemicals used on site.	COR review of submitted documentation (CHMI, SDS, Usage Forms). COR [REDACTED]

TECHNICAL EXHIBIT 2

Deliverables Schedule

<u>Deliverable</u>	<u>Frequency</u>	<u># of Copies</u>	<u>Medium/Format</u>	<u>Submit To</u>
Lock Out/Tag Out (LOTO) Procedure PWS paragraph 1.13.1	One month prior to first quarterly maintenance visit and if there are any revisions	Initial electronic copy and each time there is a change to the LOTO	PDF or WORD	COR david.otero1.civ@us.navy.mil
PSNS&IMF 5090/132 CHMI Form PWS paragraph 1.13.2	One month prior to first quarterly maintenance visit and every time there is a change to the chemicals that will be used for maintenance	Initial electronic copy and each time there is a change to the chemicals being used to perform work according to PWS	PDF or WORD	COR david.otero1.civ@us.navy.mil
Safety Data Sheets PWS paragraph 1.13.2	One month prior to first quarterly maintenance visit and every time there is a change to the chemicals that will be used for maintenance	Initial electronic copy and each time there is a change to the chemicals being used to perform work according to PWS	PDF or WORD	COR david.otero1.civ@us.navy.mil
PSNS Receipt and Monthly Usage Form PWS paragraph 1.13.3	Upon completion of each quarterly maintenance visit	Electronic copy each maintenance visit	PDF or WORD	COR david.otero1.civ@us.navy.mil
Service Report PWS paragraph 5.2	Upon completion of each scheduled maintenance and corrective maintenance	Electronic copy each maintenance visit	PDF or WORD	COR david.otero1.civ@us.navy.mil
Repair Quote PWS paragraph 5.3	Upon completion of troubleshoot services	Electronic copy each troubleshoot visit	PDF or WORD	COR david.otero1.civ@us.navy.mil

<u>Deliverable</u>	<u>Frequency</u>	<u># of Copies</u>	<u>Medium/Format</u>	<u>Submit To</u>